

Machine Learning–Based Estimation of Wireless Coverage Maps Using Terrain Data and GPS-Sampled Signal Measurements

Gino MARTELLI, Luka SUTKOVIC, Didier AUROUX, Van Lic TRAN

Abstract—Accurate radio coverage estimation is essential for the deployment and optimization of Low Power Wide Area Networks (LPWANs) such as LoRaWAN. This paper presents a data-driven framework for RSSI prediction that combines real-world LoRaWAN measurements with geographic information system (GIS) data. A dataset of 10,203 real-world LoRaWAN RSSI measurements collected in Da Nang, Vietnam, was used to train and evaluate the proposed approach. The model incorporates radio transmission parameters, geometric and terrain-related features, land-use information, and nearest-neighbor signal descriptors extracted from a spatial reference database. Several machine learning algorithms were evaluated, and an optimized Extra Trees regressor was selected as the final model. The proposed method achieved an average coefficient of determination of $R^2 = 0.9285$, a mean absolute error of 1.93 dBm, and a root mean square error of 2.92 dBm. The trained model was integrated into an inference engine capable of predicting RSSI at arbitrary geographic locations, enabling accurate wireless coverage mapping and supporting LoRaWAN network planning.

Index Terms—RSSI, Wireless Coverage Estimation, Machine Learning, LoRaWAN, LPWAN, Terrain-Aware Propagation

I. INTRODUCTION

Low Power Wide Area Networks (LPWANs) have become a key communication technology for Internet of Things (IoT) applications such as smart cities, environmental monitoring, and industrial automation. Among LPWAN technologies, LoRaWAN offers long-range communication with low power consumption, making accurate radio coverage prediction essential for gateway deployment and network planning [1], [2].

Radio propagation prediction remains challenging because signal attenuation depends not only on the transmitter–receiver distance but also on terrain, vegetation, buildings, land use, and other environmental factors [3], [4]. Traditional propagation models such as Okumura–Hata, COST-231, ECC-33, and the Irregular Terrain Model (ITM) are computationally efficient but often fail to accurately represent complex environments, leading to significant prediction errors outside their calibration conditions [5]–[10].

Machine learning has recently emerged as an effective alternative by learning nonlinear propagation behaviors directly from measurement data [11]–[15]. However, many existing approaches rely on simulated datasets, require computationally

intensive geographic processing, or overlook practical deployment aspects required for real-time coverage mapping [16], [17].

To address these limitations, this paper proposes a terrain-aware and neighbor-assisted machine learning framework for LoRaWAN RSSI prediction. The proposed approach combines radio parameters, geometric descriptors, terrain and Fresnel-zone characteristics, land-use information, and neighboring signal statistics extracted from a spatial reference database. The model is trained using 10,203 real-world RSSI measurements collected in Da Nang, Vietnam, and employs an optimized Extra Trees regressor as the prediction engine.

The main contributions of this work are fourfold. First, a real-world LoRaWAN RSSI dataset comprising 10,203 measurements collected using 13 gateways was constructed. Second, a feature engineering framework integrating radio, geometric, terrain, land-use, obstruction, Fresnel-zone, and neighbor-based descriptors was developed. Third, an optimized Extra Trees model achieved an R^2 of 0.9285, a MAE of 1.93 dBm, and a RMSE of 2.92 dBm. Finally, an inference engine was designed for practical integration into wireless coverage mapping and network planning applications.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the proposed methodology. Section IV describes the experimental setup. Section V discusses the results, and Section VI concludes the paper.

II. RELATED WORK

Wireless coverage prediction is a fundamental problem in radio network planning and optimization. Coverage estimation has traditionally relied on empirical propagation models and interpolation techniques, but increasingly employs data-driven machine learning methods as heterogeneous environments make accurate modeling more challenging.

A. Traditional Propagation Models

Early coverage prediction was based on empirical models derived from measurement campaigns. Among the most widely used are the Okumura and Hata models for urban, suburban, and rural environments. Later extensions, including COST-231 Hata and COST-231 Walfisch–Ikegami, extended the frequency range and incorporated dense urban morphology and non-line-of-sight (NLOS) propagation effects [5]–[7].

Other widely used models include the Irregular Terrain Model (ITM or Longley–Rice) and ECC-33, which account for terrain and diffraction effects [3], [8]. Although computationally efficient, empirical models often perform poorly outside their calibration environments, with reported prediction errors ranging from 10 to more than 40 dB in complex scenarios because they inadequately capture terrain, vegetation, and urban obstructions [4], [18].

Several studies evaluated these models for LoRaWAN. Petäjäjärvi et al. [1] showed that LoRa communication can exceed 15 km under favorable conditions, while El Chall et al. [2] and Callebaut and Van der Perre [19] demonstrated that environment-specific propagation models outperform conventional empirical formulations.

B. Terrain-Aware and Measurement-Based Coverage

To improve prediction accuracy, several works incorporated terrain and geographic information into propagation modeling. Demetri et al. [18] combined remote sensing with propagation models to reduce prediction errors, while Bezerra et al. [9] showed that no single empirical model consistently performs well across deployment scenarios.

Other studies highlighted the influence of terrain morphology, vegetation, weather, and Fresnel-zone obstruction on LoRa performance [20], [21]. Lima et al. [10] further developed dedicated propagation models for Amazonian environments, outperforming conventional Okumura-Hata formulations. Nevertheless, terrain-aware empirical models remain dependent on environment-specific calibration.

C. Machine Learning for Coverage and Path-Loss Prediction

Machine learning has become an effective alternative by learning nonlinear relationships between signal measurements, network parameters, and environmental features [11], [12]. Alimpertis et al. [16] generated city-scale signal-strength maps using Random Forests, while Moraitis et al. [22] showed that ensemble methods outperform conventional regression techniques for rural path-loss prediction. Fauzi et al. [13] compared several supervised learning algorithms for cellular coverage prediction, identifying Random Forest and Gaussian Process Regression as the best trade-off between accuracy and complexity. Similarly, He et al. [23] proposed a Random Forest path-loss model, and the MLOE framework [17] demonstrated practical deployment of machine learning for network planning.

For LPWANs, Hosseinzadeh et al. [24] introduced an ANFIS-based LoRaWAN propagation model, while their later work [14] showed that explainable ensemble methods achieve accurate predictions while providing feature interpretability. González-Palacio et al. [25] jointly modeled path loss and shadow fading using machine learning to improve LoRaWAN energy efficiency.

D. Deep Learning and Hybrid Learning Approaches

Deep learning methods have recently emerged for wireless coverage prediction. Zeng et al. [15] demonstrated that

combining Deep Neural Networks with semantic segmentation improves prediction accuracy using image-based environmental representations, although at the cost of larger datasets and higher computational requirements.

Hybrid approaches combining empirical models with machine learning have also shown promising results. Alobaidy et al. [26] combined Free-Space Path Loss with several learning algorithms, achieving up to $R^2 = 0.89$ on more than 43,000 measurements while outperforming ITM, COST-231 Hata, and ECC-33.

Overall, the literature shows a clear transition from empirical propagation models toward data-driven machine learning and deep learning methods, which differ substantially in their input features, computational complexity, and deployment objectives. Table I summarizes representative approaches and their limitations.

E. Research Gap and Contribution

As summarized in Table I, empirical propagation models remain attractive because of their simplicity but often degrade in heterogeneous environments. Machine learning approaches generally provide higher accuracy, yet many focus on cellular networks, rely on computationally intensive architectures or simulated datasets, or exploit only a limited subset of environmental information.

This work addresses these limitations by combining radio parameters, geometric descriptors, terrain statistics, obstruction and Fresnel-zone metrics, land-use information, and neighboring RSSI features within an Extra Trees regression framework. Trained on more than 10,000 real-world LoRaWAN measurements collected in Da Nang, Vietnam, the proposed model aims to provide accurate RSSI prediction while remaining suitable for practical wireless coverage mapping and network planning.

III. METHODOLOGY

A. Overall System Architecture

The proposed framework is organized into three main stages: data acquisition, offline model training, and online prediction. Figure 1 illustrates the overall architecture.

The data acquisition stage combines LoRaWAN measurements with external geographic information. The latter includes Digital Elevation Model (DEM) data used to derive terrain-related features and OpenStreetMap (OSM) data used to extract land-use information. These datasets are processed to build the measurement dataset used for model training, while terrain information is also stored in a local database for later reuse during inference.

During deployment, the prediction engine receives a query location together with the radio configuration. It then loads the trained Extra Trees model, the gateway database, the reference measurement database, and the local terrain database. The required features are generated on demand, including geometric, terrain, land-use, and nearest-neighbor descriptors, before estimating the RSSI. Repeating this process over multiple locations enables the generation of wireless coverage maps.

TABLE I
COMPARISON OF EXISTING WIRELESS COVERAGE PREDICTION APPROACHES AND IDENTIFIED RESEARCH GAPS

Study	Method	Advantages	Limitations / Research Gap
Okumura-Hata, COST-231, ECC-33, ITM [5]–[8]	Empirical propagation models	Simple, computationally efficient, widely used	Require environment-specific calibration and poorly capture heterogeneous terrain, vegetation, and urban obstructions.
Demetri et al. [18]	Remote sensing-assisted LoRa coverage	Integrates geographic information	Still relies on propagation-model assumptions.
Bezerra et al. [9]	ITM, ITWOM, Okumura-Hata evaluation	Highlights terrain-aware modeling	No single model performs consistently across environments.
El Chall et al. [2]	Environment-specific LoRaWAN models	Improves prediction over standard models	Location-dependent; requires dedicated calibration.
Lima et al. [10]	LoRa models for Amazon regions	Effective in dense vegetation	Limited generalization.
Alimpertis et al. [16]	Random Forest signal mapping	Large-scale ML coverage prediction	Focuses on cellular networks with limited terrain descriptors.
Fauzi et al. [13]	LR, ANN, SVM, RF, GPR	Compares multiple supervised ML models	Limited to cellular networks.
MLOE [17]	Random Forest coverage estimator	Fast and deployment-oriented	Limited environmental features.
Hosseinazadeh et al. [24]	ANFIS LoRaWAN model	Captures nonlinear propagation	Higher complexity and training requirements.
Hosseinazadeh et al. [14]	Explainable ML	Prediction with feature interpretation	Focuses on link-budget analysis.
González-Palacio et al. [25]	ML path-loss + shadowing	Improves propagation and energy efficiency	Designed mainly for power optimization.
Zeng et al. [15]	DNN + semantic segmentation	High prediction accuracy	Requires large datasets and high computational cost.
Alobaidy et al. [26]	Hybrid stacked ML	Accurate with good generalization	Higher model complexity.
This Work	Extra Trees with GIS and spatial features	Combines radio, terrain, land-use, Fresnel, obstruction, and neighboring RSSI features.	Requires representative reference measurements; GIS feature extraction dominates inference time.

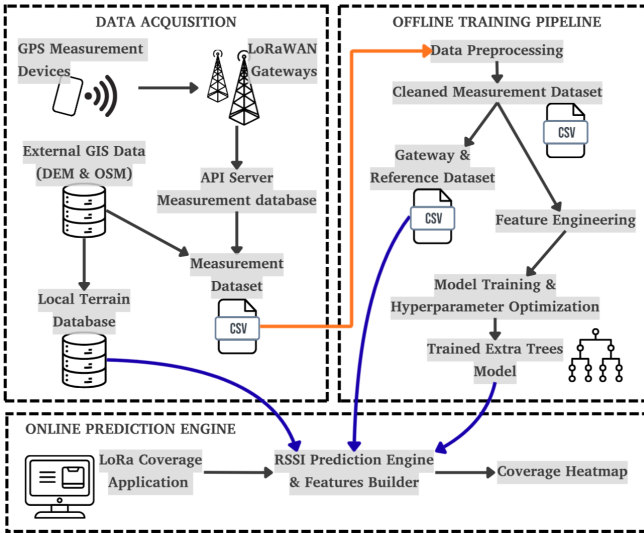


Fig. 1. Overall architecture of the proposed LoRaWAN RSSI prediction framework.

B. Data Processing and Feature Engineering

The proposed framework combines raw LoRaWAN measurements with external geographic datasets. Before feature extraction, invalid records, including measurements with missing or erroneous geographical coordinates, are removed and obvious formatting errors in latitude and longitude values are automatically corrected.

The cleaned measurements are enriched with terrain elevation from the Shuttle Radar Topography Mission (SRTM) [27]

and land-use information from OpenStreetMap. The resulting feature set captures three complementary aspects of radio propagation: radio configuration, geometric and environmental characteristics, and local spatial information extracted from neighboring measurements.

1) *Radio Features*: Only parameters available before transmission are considered, namely the operating frequency and the spreading factor.

2) *Geometric Features*: The relative gateway–receiver geometry is described using the three-dimensional distance and its logarithm, latitude and longitude offsets, the azimuth angle, gateway and receiver elevations, elevation difference, and elevation angle. The logarithmic distance is included since path loss approximately follows a logarithmic relationship with distance.

3) *Terrain Features*: Terrain descriptors are extracted from the SRTM digital elevation model, while land-use information is obtained from OpenStreetMap.

For each gateway–receiver path, the terrain profile is summarized using elevation statistics (mean, minimum, maximum, range, and standard deviation), average terrain slope, terrain roughness, maximum obstruction, Fresnel obstruction ratio, minimum and mean Fresnel clearance, and the residential land-use ratio. These descriptors capture diffraction, shadowing, and environmental attenuation effects.

a) *Terrain Descriptor Computation*: The local terrain slope is estimated from a 3×3 DEM window centered on the receiver location:

$$S = \frac{1}{N} \sum_{i=1}^N \sqrt{g_{x,i}^2 + g_{y,i}^2}. \quad (1)$$

where g_x and g_y denote the horizontal and vertical elevation gradients obtained through finite differences. Terrain roughness is defined as the standard deviation of elevations within a local 5×5 DEM neighborhood, where larger values indicate more irregular terrain.

b) *Line-of-Sight and Obstruction Features*: Terrain elevations are sampled every 30 m along each gateway–receiver path. The line-of-sight (LOS) height is obtained by linear interpolation between the gateway and receiver antenna heights:

$$h_{\text{LOS}}(i) = h_g + \frac{i}{N-1}(h_r - h_g). \quad (2)$$

The terrain obstruction is then defined as

$$O_i = h_{\text{terrain}}(i) - h_{\text{LOS}}(i), \quad (3)$$

and the maximum obstruction as

$$O_{\max} = \max_i O_i, \quad (4)$$

where positive values indicate terrain intersecting the direct propagation path.

The first Fresnel zone radius is computed as

$$F_1 = \sqrt{\frac{\lambda d_1 d_2}{d_1 + d_2}}, \quad (5)$$

where λ is the wavelength and d_1 and d_2 are the distances from the considered point to the transmitter and receiver, respectively. The Fresnel clearance is defined as

$$C_i = h_{\text{LOS}}(i) - F_1(i) - h_{\text{terrain}}(i). \quad (6)$$

The Fresnel obstruction ratio corresponds to the proportion of sampled points for which $C_i < 0$. This descriptor quantifies the fraction of the propagation path affected by Fresnel-zone blockage.

C. Nearest-Neighbor Spatial Features

A key contribution of the proposed framework is the integration of local historical measurements as predictive features. Rather than directly applying K-nearest-neighbor (KNN) interpolation, neighboring observations are summarized into statistical descriptors that are used as additional inputs to the machine learning model.

1) *Neighbor Selection*: Historical measurements are grouped by gateway and indexed using gateway-specific KD-Trees, enabling efficient nearest-neighbor retrieval.

Given a query point, the corresponding gateway KD-Tree is first queried to retrieve a set of candidate neighbors. Measurements located within a minimum radius

$$d < d_{\min} \quad (7)$$

are discarded in order to avoid duplicated samples collected at the same physical position. The remaining candidates are ranked according to

Algorithm 1 Training procedure

Require: Measurement dataset $\mathcal{D}$

for each cross-validation fold **do**

Split $\mathcal{D}$ into training and validation sets

Build gateway-specific KD-Trees using training samples only

Generate neighbor features for training samples

Generate neighbor features for validation samples using the training KD-Trees

Train Extra Trees model

Evaluate R^2 , MAE and RMSE

end for

Retrain final model on the complete dataset

Build final gateway-specific KD-Trees

Save model and auxiliary databases

$$S_i = d_i + \alpha |D_i - D_q|, \quad (8)$$

where d_i is the spatial distance to neighbor i , D_i and D_q denote the gateway distances of the neighbor and query point, respectively, and α controls the importance of gateway-distance similarity. The highest-ranked neighbors are retained for feature generation.

2) *Neighbor Statistics*: For the selected neighbors, several statistical descriptors are computed, including the RSSI, spatial distance and gateway distance of the closest neighbor, the mean and weighted mean RSSI, the RSSI standard deviation, and the mean spatial and gateway distances.

The weighted RSSI average is computed as

$$\bar{R} = \frac{\sum_i R_i e^{-d_i/\tau}}{\sum_i e^{-d_i/\tau}}, \quad (9)$$

where R_i is the RSSI of neighbor i , d_i its spatial distance, and τ is a distance-decay parameter.

D. Machine Learning Model

Algorithm 1 summarizes the training and evaluation procedure used throughout this work. During each cross-validation fold, the reference measurement database and gateway-specific KD-Trees were built exclusively from the training partition. Neighbor-based features were then generated independently for the training and validation sets using these training-only data, preventing information leakage. After cross-validation, the final model was retrained on the complete dataset and stored together with the gateway-specific KD-Trees and auxiliary databases required for deployment.

E. Inference Engine

The prediction engine was designed independently from the training framework and only requires the artifacts generated during offline training. At initialization, it loads the trained Extra Trees model together with the gateway database, gateway-specific KD-Trees, reference measurements, and terrain datasets. Algorithm 2 summarizes the online prediction procedure.

Algorithm 2 Online RSSI prediction

Require: Query location (lat, lon)

if gateway not specified **then** State Select gateway using the deployment strategy

end if

if reference measurement exists within 30 m **then**

return measured RSSI

end if

if query location is outside the gateway operating range **then**

return -120 dBm

end if

Generate geometric, terrain and neighbor features

Predict RSSI using the trained model **return** RSSI estimate

During inference, geometric, terrain, and neighbor-based features are generated dynamically before being passed to the trained model. Several deployment optimizations are also implemented, including automatic gateway selection, direct reuse of nearby reference measurements when available, and rejection of predictions outside the observed gateway coverage. These optimizations improve inference efficiency without affecting the training or evaluation procedure.

IV. EXPERIMENTAL SETUP

A. Dataset

The proposed framework was evaluated using a dataset of 10,203 cleaned LoRaWAN RSSI measurements collected in Da Nang, Vietnam. Measurements were acquired using 13 LoRaWAN gateways and three end devices. Each record contains the transmitter geographical coordinates, the associated gateway, the operating frequency, the spreading factor, and the measured RSSI. During preprocessing, a small number of duplicated measurements collected at identical locations exhibited physically inconsistent RSSI values. Only the most extreme outliers were removed while preserving the remaining observations.

TABLE II
RSSI STATISTICS OF THE FINAL DATASET

Statistic	RSSI (dBm)
Minimum	-135
25th percentile	-114
Median	-110
Mean	-108.44
75th percentile	-105
Maximum	-28
Standard deviation	10.96

Table II summarizes the RSSI distribution of the final dataset. Measurements were collected using three spreading factors (SF7, SF10, and SF12) and several operating frequencies within the Vietnamese LoRaWAN band. Their distributions are reported in Tables III and IV.

B. Cross-Validation Protocol

Model performance was evaluated using shuffled five-fold cross-validation, with 80% of the data used for training and

TABLE III
DISTRIBUTION OF SPREADING FACTORS

Spreading Factor	Percentage (%)
SF7	27.72
SF10	24.45
SF12	47.83

TABLE IV
FREQUENCY DISTRIBUTION

Frequency (MHz)	Percentage (%)
921.4	43.84
921.6	3.09
921.8	1.08
922.0	0.98
922.2	48.79
922.4	0.90
922.6	0.73
922.8	0.60

20% for validation in each fold. To prevent information leakage, the reference measurement database used to generate nearest-neighbor features was reconstructed independently for each training fold, ensuring that validation samples were never used during feature generation. To verify that the random five-fold partitioning does not introduce sampling bias, we compared the distributions of RSSI and features between the training and testing subsets using the two-sample Kolmogorov–Smirnov test. No significant differences were observed across the folds.

C. Model Configuration

The final prediction model is based on the Extra Trees regressor. After hyperparameter optimization, the model was configured with 650 trees, a maximum depth of 18, a minimum of 10 samples required for each split, and a maximum feature ratio of 0.7.

The nearest-neighbor module used $K = 9$, $K_{search} = 11$, a minimum distance threshold of 0.1 m, a gateway-distance weighting coefficient $\lambda = 1.1$, and a distance-decay parameter $\tau = 30$ m for the weighted RSSI average. These parameters provided the best trade-off between prediction accuracy and computational complexity.

D. Evaluation Metrics

The prediction performance was evaluated using three standard regression metrics: the coefficient of determination (R^2), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). These metrics provide complementary information on the quality of the predictions. The MAE measures the average magnitude of the prediction errors, the RMSE penalizes larger errors more heavily due to the squared term, and the coefficient of determination (R^2) quantifies the proportion of the variance in the measured RSSI values explained by the model.

V. RESULTS AND DISCUSSION

A. Performance of Machine Learning Models

Several supervised regression algorithms were evaluated using the experimental protocol described in Section IV. Table V summarizes their predictive performance.

TABLE V
PERFORMANCE COMPARISON OF EVALUATED MACHINE LEARNING MODELS

Model	MAE (dBm)	RMSE (dBm)	R^2
SVR	2.79	5.49	0.7610
MLP Regressor	2.58	4.19	0.8609
HistGradientBoosting	2.22	3.72	0.8904
XGBoost	2.14	3.80	0.8857
Random Forest	2.11	3.42	0.9073
Extra Trees	2.03	3.45	0.9056
Final Proposed Extra Trees	1.93	2.92	0.9285

Tree-based ensemble methods achieved the best results, clearly outperforming SVR, MLP, and boosting approaches. Random Forest and Extra Trees produced similar performance, with Extra Trees providing slightly lower MAE and RMSE.

A direct comparison between both ensemble methods using the final feature set is presented in Table VI. Extra Trees consistently achieved the best overall performance and was therefore selected as the final prediction model.

TABLE VI
PERFORMANCE COMPARISON OF TREE-BASED ENSEMBLE MODELS

Model	MAE (dBm)	RMSE (dBm)	R^2
Random Forest	2.05	3.24	0.9117
Extra Trees	2.00	3.21	0.9135
Final Proposed Extra Trees	1.93	2.92	0.9285

The final optimized model achieved an average cross-validation performance of $R^2 = 0.9285 \pm 0.0065$. Based on the absolute prediction errors obtained over all cross-validation predictions, the estimated 95% confidence interval for the MAE is 95% CI = [1.89, 1.97] dBm, indicating a stable estimate of the average prediction error.

Figure 2 compares predicted and measured RSSI values. Most samples lie close to the identity line, indicating good agreement across the full RSSI range.

B. Coverage Map Generation

Figure 3 illustrates RSSI prediction over Da Nang on a regular 50 m grid. The left panel shows the resulting RSSI coverage map, where spatial variations in signal strength are captured across urban areas and surrounding regions. The right panel depicts the gateway assignment used by the inference engine for each prediction location. When no gateway is explicitly provided, the system selects the gateway associated with the nearest reference measurement, rather than simply choosing the geographically closest gateway. This design choice results in spatially varying gateway regions, where adjacent areas may be associated with different gateways depending on the underlying measurement distribution. These transitions reflect the actual network attachment behavior observed in the dataset rather than purely geometric proximity. The reliability of

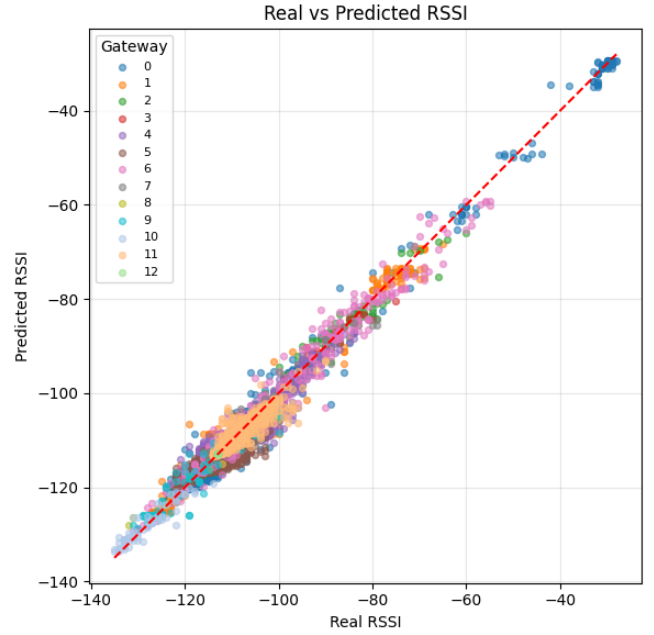


Fig. 2. Predicted RSSI versus measured RSSI. The dashed line represents the ideal prediction ($y = x$).

predictions depends on the spatial density of the available measurements. Regions with sparse coverage may exhibit higher uncertainty in the estimated RSSI values.

C. Impact of Feature Engineering

Prediction accuracy improved progressively as additional feature groups were introduced. The largest gain resulted from incorporating nearest-neighbor descriptors, particularly the weighted neighbor RSSI and the RSSI of the closest reference measurement.

Several candidate variables, including redundant geometric and path-based descriptors, were discarded because they did not improve cross-validation performance. The final feature set combines radio, geometric, terrain, Fresnel, land-use, and neighbor-based information.

Feature importance was assessed using permutation importance on the validation folds. Each feature was randomly shuffled while all others remained unchanged, and its importance was defined by the resulting degradation in prediction performance. Importance scores were averaged over the five cross-validation folds using ten permutations per feature.

Table VII presents the ten most influential variables.

Neighbor-based RSSI descriptors rank among the most informative variables, confirming the value of local historical measurements. Terrain-related variables also contribute substantially, while Fresnel and obstruction descriptors exhibit relatively low individual importance but provide complementary information, as their removal degraded prediction accuracy during feature selection.

D. Error Analysis

Figure 4 shows the evolution of the absolute prediction error as a function of the transmitter-gateway distance. For

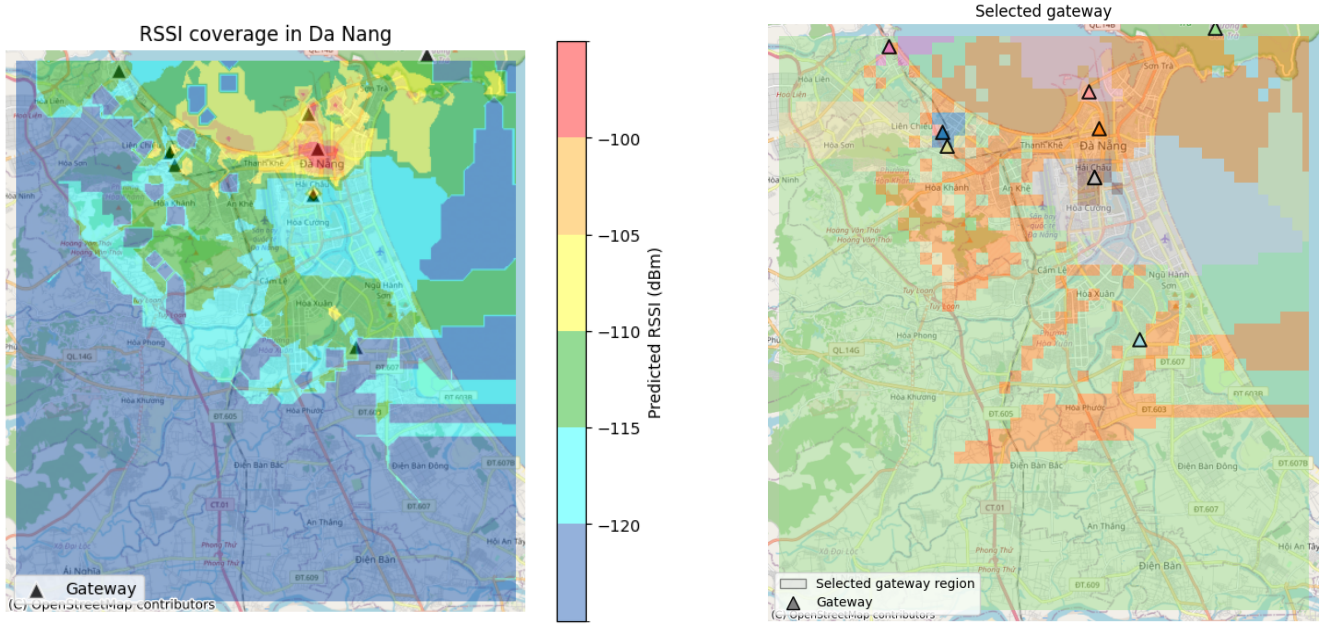


Fig. 3. (Left) Predicted RSSI coverage map over Da Nang generated on a 50 m grid. (Right) Gateway assignment per prediction location based on nearest-reference-measurement selection.

TABLE VII
TOP 10 MOST IMPORTANT FEATURES

Feature	Importance
Residential ratio	0.159
Gateway ID	0.119
Neighbor RSSI weighted mean	0.083
Frequency	0.079
Neighbor RSSI mean	0.061
RSSI closest point	0.059
Elevation angle	0.047
Spreading factor	0.034
Log distance (3D)	0.034
Elevation	0.031

each distance bin, the median error and interquartile range are reported. To quantify the data density across propagation distances, the dataset was divided into three distance ranges: < 5 km, $5\text{--}10$ km, and > 10 km. These ranges contain 7,381 (72.3%), 1,565 (15.3%), and 1,257 (12.3%) measurements, respectively. While the dataset is naturally denser at shorter distances, the long-distance range still includes more than 1,200 samples, suggesting that the corresponding statistics remain reasonably reliable despite the lower sampling density.

Overall, the results indicate that the prediction error does not systematically increase with distance. Although the dataset contains a higher density of samples at short ranges due to the measurement campaign, the model maintains stable performance across different distance regimes. This suggests that the learned representation generalizes beyond local spatial neighborhoods and is not solely driven by proximity effects in the training data.

Slight increases in variability can be observed in specific distance intervals, which may be associated with heterogeneous propagation conditions and local measurement inconsistencies rather than a monotonic degradation of model

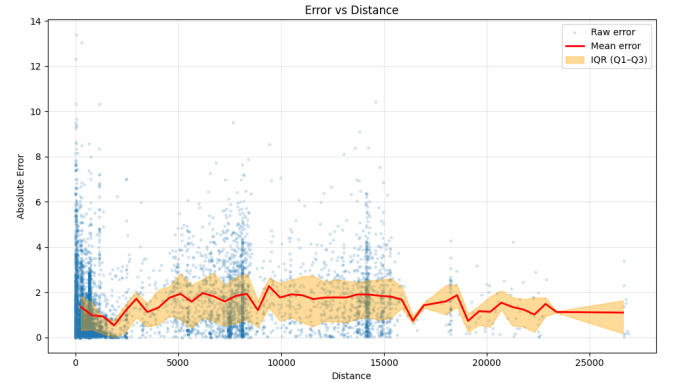


Fig. 4. Absolute prediction error as a function of transmitter-gateway distance. The curve shows median error and interquartile range per distance bin.

performance with distance.

The largest residuals were mainly associated with duplicated measurements acquired at identical locations but exhibiting substantially different RSSI values, likely due to interference, hardware variability, or temporary environmental conditions.

Only the most extreme outliers were removed, improving the average cross-validation score from $R^2 = 0.9257$ to 0.9285 while preserving nearly all measurements.

The maximum remaining prediction error is approximately 13 dB, with most predictions exhibiting substantially smaller deviations, as shown in Figure 5. The error distribution is approximately symmetric and centered around zero, indicating that the model does not exhibit a significant systematic tendency to overestimate or underestimate the RSSI. The highest concentration of predictions occurs near zero error with approximately 1,800 samples, while the number of observations

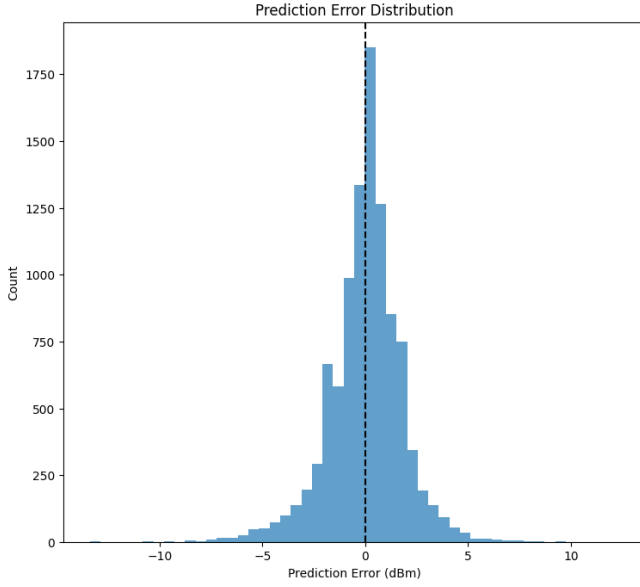


Fig. 5. Distribution of prediction errors on the evaluation dataset.

decreases progressively as the absolute error increases.

E. Independent Field Validation

An independent measurement campaign comprising approximately 800 unseen LoRaWAN measurements was used to assess generalization. Approximately 600 measurements were received by a gateway absent from the training dataset. In these cases, the inference engine selected the gateway associated with the nearest reference measurements, located about 190 m from the actual gateway, and computed all geometry- and terrain-related features relative to this surrogate gateway.

The proposed framework achieved a MAE of 3.31 dBm and an RMSE of 4.60 dBm. Although higher than the cross-validation error, these results indicate that the model remains reasonably robust to moderate gateway location mismatches.

F. Practical Deployment Considerations

Unlike many machine learning studies that focus exclusively on prediction accuracy, this work also addresses deployment constraints. The final prediction engine is completely separated from the training framework. During initialization, only the artifacts produced during offline training are loaded into memory, including the trained model, gateway database, reference measurements, and terrain datasets. Feature generation is performed on demand for arbitrary GPS coordinates, after which RSSI predictions are produced directly by the trained model. No retraining, dataset generation, or preprocessing operations are required. This architecture enables straightforward integration into interactive wireless coverage mapping applications. Since the trained model is loaded only once, repeated predictions can be generated without retraining or rebuilding the spatial indexes. The main computational cost is associated with feature generation, particularly the extraction of terrain-related descriptors.

G. Comparison with Existing Methods

Table VIII compares the proposed approach with representative machine-learning-based wireless coverage prediction methods. The proposed model achieves competitive performance ($R^2 = 0.9285$, RMSE = 2.92 dB), outperforming recent LoRaWAN approaches while remaining comparable to state-of-the-art cellular solutions despite relying on a simpler Extra Trees model and engineered GIS features.

H. Limitations

Although the proposed framework achieves high prediction accuracy, several limitations should be acknowledged.

First, the measurement campaigns were conducted along predefined routes rather than uniformly covering the study areas. Consequently, measurements are spatially clustered, with many samples collected only a few dozen meters apart. Although nearest-neighbor descriptors are generated without information leakage between training and validation folds, this spatial sampling pattern may still favor the learning of highly localized propagation characteristics. As a result, the reported performance should be interpreted in the context of the geographical distribution of the available measurements.

Second, the proposed approach relies on external geographic information, including Digital Elevation Models (DEM) and OpenStreetMap land-use data, to construct terrain-related descriptors. Applying the framework to a new geographical region therefore requires the availability of equivalent GIS datasets so that the same feature extraction process can be reproduced.

Finally, most of the computational cost during deployment is associated with feature generation rather than machine learning inference itself. In particular, obstruction and Fresnel-zone descriptors require sampling the propagation path every 30 m between the gateway and the query location, which dominates the inference time when generating dense coverage maps. While the trained Extra Trees model predicts RSSI values in a negligible amount of time, feature extraction represents the dominant computational cost when generating large-scale coverage maps.

VI. CONCLUSION

This paper presented a machine-learning-based RSSI prediction framework for LoRaWAN coverage estimation. The proposed approach combines radio configuration parameters, geometric descriptors, terrain characteristics, Fresnel-zone indicators, land-use information, and nearest-neighbor measurement statistics within a unified Extra Trees regression model. Evaluated on a dataset of more than 10,000 real-world measurements collected from 13 gateways, the proposed model achieved an average cross-validation performance of $R^2 = 0.9285$, a MAE of 1.93 dBm, and a RMSE of 2.92 dBm. Feature importance analysis further highlighted the complementary contributions of environmental descriptors and neighboring measurements to prediction accuracy.

Beyond predictive performance, the trained model was integrated into a dedicated inference engine capable of estimating RSSI from geographical coordinates only, making it suitable

TABLE VIII
COMPARISON WITH REPRESENTATIVE MACHINE LEARNING-BASED WIRELESS COVERAGE PREDICTION METHODS

Reference	Tech.	Model	RMSE	R^2	Remarks
[14]	LoRaWAN	RF	5.70	0.77	Link-budget modeling with fewer environmental features.
[26]	LoRa	Hybrid ML	4.77	0.89	Hybrid stacked model for path-loss prediction rather than RSSI mapping.
[17]	LTE	RF	2.65	0.93	Cellular coverage prediction using radio-network features.
[15]	Cellular	DNN	1.97	–	Deep learning with semantic segmentation and image-based features.
This Work	LoRaWAN	Extra Trees	2.92	0.9285	GIS-derived terrain, Fresnel, land-use, and neighbor RSSI features.

for wireless coverage mapping and network planning applications. An independent measurement campaign involving approximately 800 previously unseen measurements, including data acquired through a previously unseen gateway, confirmed the robustness of the proposed framework under realistic deployment conditions, yielding a MAE of 3.31 dBm.

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